



EmoGlasses: Emotion-Sensing Smart Glasses for EEG-Based Human-Environment Interaction

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This project aims to develop a smart glasses system that detects users' emotional states using EEG signals and enables adaptive interaction with smart environments.



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Motivation & Background

Many existing BCI devices are bulky, uncomfortable, expensive and invasive.

EEG-based emotion recognition has shown potential in supporting adaptive human-computer interaction, especially for users with emotional regulation difficulties or special needs.

Glasses are natural, non-intrusive wearables that blend into everyday routines—making them perfect for convenient and effective EEG sensing.



System Concept Design

EEG Signal Acquisition

Dry electrodes are embedded on the sides of smart glasses, positioned around F7/T3 and F8/T4, with probe details shown later. These regions cover parts of the prefrontal and temporal lobes, which are associated with emotion processing and facial muscle activity.

EEG signals are captured using the PiEEG board connected to Raspberry Pi 5.

Signal Preprocessing and Multimodal Input Generation

On-device preprocessing on the Pi includes filtering (1–40 Hz), normalization, and segmentation by task type.

Each EEG trial is then transformed into three modalities:

- Time-domain signal (raw EEG waveforms)
- Frequency-domain features (e.g., band power in alpha, beta, and theta bands)
- Image representation (e.g., spectrogram)

Multimodal Deep Learning Model

Three parallel neural networks extract features from each modality. The outputs are fused via a late fusion layer for final classification.

The model predicts both emotional states (positive, neutral, negative) and user commands (e.g., left/right gaze or jaw clench).

System Response

Emotion detection is used to initiate personalized prompts or suggestions.

Intent detection (e.g., eye movement or jaw clench) triggers specific commands like UI navigation or smart device control.

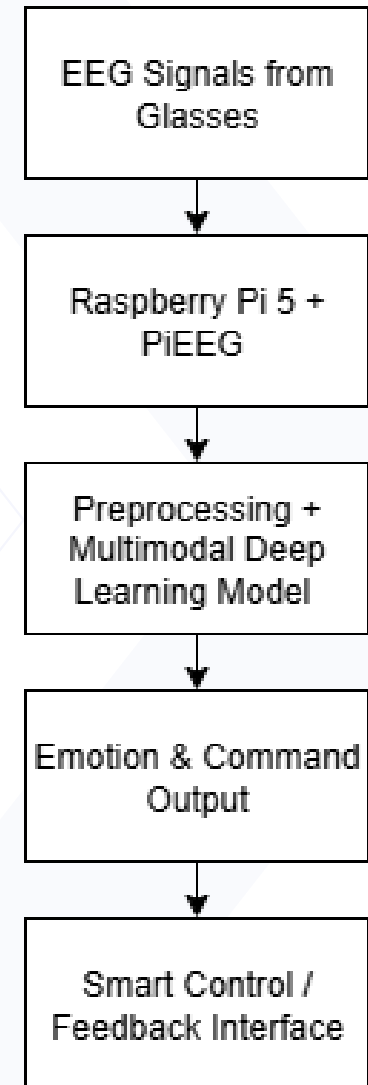


Fig1. System Workflow Overview



Electrode Placement & Signal Types

Dry electrodes are embedded into the glasses' side frames, approximately over the F7–T3 and F8–T4 areas.

Mapped EEG Signal Types

F7 / F8: Located at the prefrontal cortex, these electrodes capture emotion-related EEG signals (e.g., increased alpha-band activity (8–13 Hz) is commonly observed during relaxed wakefulness, while stress or negative affect may induce alpha suppression accompanied by elevated theta or low beta band).
→ Used for real-time emotion estimation (Calm / Positive / Negative). (Zhou, F., et al., 2020)

T3 / T4: Positioned near facial muscles and temporalis, these channels are highly sensitive to jaw muscle activity (EMG) and eye movement artifacts (EOG).
→ Used for voluntary control signals: “confirm” via jaw clench and “directional choice” via eye movement. (Ebersole, J.S., et al., 2023)

These locations are specifically chosen to enable both emotional state recognition and intent-based BCI control, using only minimal dry electrodes. (Rakhmatulin, I., 2023)

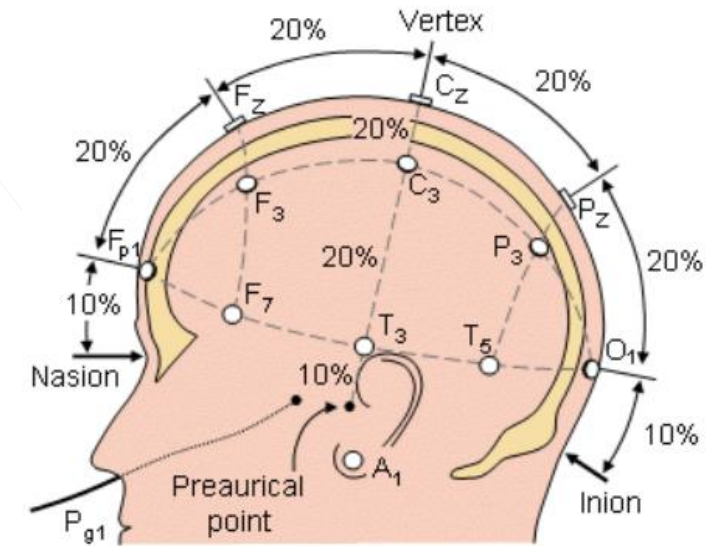


Fig2. EEG electrode placement based on the international 10–20 system. Adapted from Morley, A., Hill, L., & Kaditis, A. (2016). 10–20 system EEG Placement. European Respiratory Society.



Fig3. Prototype of 3D-Printed EEG Glasses with PiEEG and Raspberry Pi 5



Signal Acquisition & Preprocessing

PiEEG-16 Module

- Based on two ADS1299 chips, each offering 8 low-noise, 24-bit ADC channels optimized for brain-computer interface (BCI) .
- Captures microvolt-level brain signals from dry electrodes and sends digital data to the Raspberry Pi for real-time processing.

Raspberry Pi 5

- Reads digital EEG data via SPI interface
- Applies software preprocessing (e.g., 1–40 Hz bandpass filtering, z-score normalization)
- Runs lightweight CNN model for emotion classification
- Outputs predictions to UI or smart device control logic

System Summary

Together, these form a compact, affordable, and portable BCI platform for real-time EEG processing and interaction.

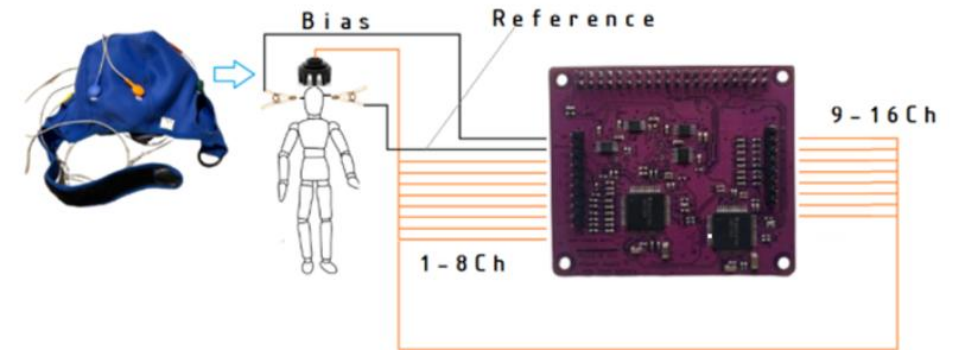


Fig4. PiEEG general schematic, Adapted from Rakhmatulin, I. (2024).

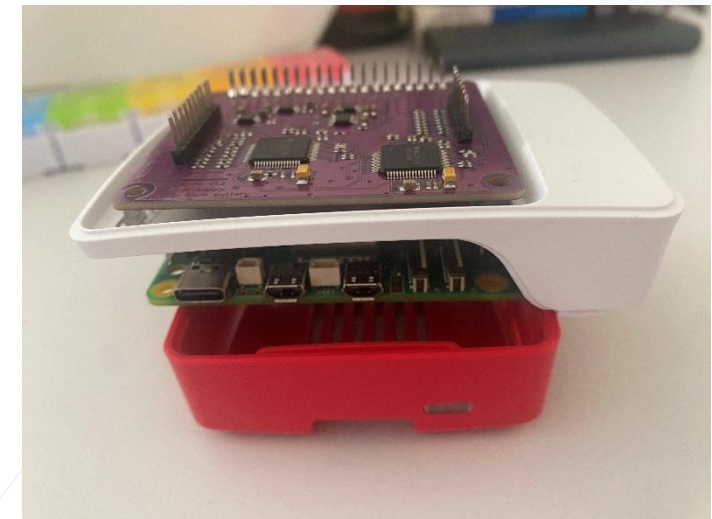


Fig5. Assembled Setup of PiEEG and Raspberry Pi5

Signal Acquisition & Preprocessing

Types of EEG Signals Captured

Signal Type	Brain Area	Description / Usage
Emotional State	F7 / F8	Prefrontal cortex – associated with emotion → Used for CNN classification (Calm, Positive, Negative)
Eye Movement	T3 / T4	Temporal lobe signals shift with horizontal eye motion → Used to detect gaze direction
Jaw Clench (Confirmation)	FT7 / FT8	Strong muscle artifacts from jaw movement → Used to confirm user intent or issue command

EEG Preprocessing Steps (for Multimodal Input)

Band-pass filtering (1–40 Hz):

Removes low-frequency drift and high-frequency noise, preserving emotion-relevant rhythms (e.g., alpha, beta).

Z-score normalization:

Standardizes each channel's amplitude across trials and subjects.

Segmentation by task type:

Emotion trials → 8–12 s EEG window

Eye movement / Jaw clench → 1–2 s window centered on the action

Multimodal Feature Extraction:

- Time-domain: Raw waveform (8 channels × N samples)
- Frequency-domain: Band power (e.g., alpha, beta) via FFT or Welch method
- Time–frequency image: Spectrogram (e.g., STFT), stored as grayscale image

Final format:

Each trial is saved as a set of synchronized features from all three domains, ready for multimodal deep learning.



Fig6. Real-Time 8-Channel Time-Domain Signals Displayed on GUI



Deep Learning Model for EEG Signal Classification



Multimodal Input:

Time-Domain: Raw or filtered EEG waveforms (8 channels $\times$ N samples).

Frequency-Domain: Log-scaled power spectra (e.g., via FFT).

Time-Frequency Image: Spectrograms or wavelet-transformed images.

Model Structure:

Three parallel subnetworks:

- 1D-CNN + LSTM for time-domain

- 1D-CNN for frequency features

- 2D-CNN (e.g., ResNet18) for time-frequency image (Although the spectrogram-based 2D-CNN branch improves classification accuracy during offline training, it may impose higher computational cost in real-time applications. If performance is acceptable, this input can be optionally disabled to ensure faster, lightweight inference.)

Fusion Strategy:

Late Fusion: Each subnetwork outputs a feature vector $\rightarrow$ concatenated $\rightarrow$ final softmax classification.

Task: Multi-class (e.g., Calm / Positive / Negative / Eye Move / Jaw Clench)

Training Info (optional if space):

Optimizer: Adam

Loss: Categorical Cross Entropy

Validation Split: e.g., 20%

Epochs: ~50 (early stop enabled)

Code and data available at

(access requires permission currently) :

https://github.com/GanshinW/emo_bci



Applications & Future Directions

After signal classification:

Emotion triggers gentle prompts

Smart home commands triggered by eye movement or EEG signal threshold detection

Basic UI interactions through lightweight system

Future Directions:

Switch from local wired setup to Wi-Fi streaming and remote server processing

Unity-based 3D interface for immersive feedback

Integration with MR or AI glasses for hands-free interaction

Add ear-region EEG sensors (e.g., near T7/T8) for discrete emotion or intent detection and extend to rear-mounted probes (e.g., O1) to enable Steady-State Visual Evoked Potential (SSVEP) based control.



References

- Zhou, F., Li, J., Zhao, W., Xu, L., Zheng, X., Fu, M., ... & Becker, B. (2020).
Empathic pain evoked by sensory and emotional-communicative cues share common and process-specific neural representations.
Frontiers in Behavioral Neuroscience, 14, 554147.
<https://doi.org/10.3389/fnbeh.2020.554147>
- Ebersole, J.S., Husain, A.M., Nordli Jr, D.R., & Kaplan, P.W. (Eds.). (2023).
Electroencephalography (EEG) - Technique and Interpretation.
Medscape / eMedicine.
<https://emedicine.medscape.com/article/1140247-overview>
- Rakhmatulin, I. (2024).
JNEEG: Jetson Nano EEG Shield for Real-Time Brain Signal Processing and Deep Learning Applications.
arXiv preprint, arXiv:2405.09575.
<https://arxiv.org/abs/2405.09575>
- Morley, A., Hill, L., & Kaditis, A. (2016).
10–20 system EEG placement. European Respiratory Society.
- Rakhmatulin, I. (2024).
PiEEG-16 to Measure 16 EEG Channels with Raspberry Pi for Brain-Computer Interfaces and EEG Devices.
arXiv preprint arXiv:2409.07491.
<https://arxiv.org/abs/2409.07491>



EmoGlasses

**Thank you for
your attention!**

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